

Repsol Tensión Centauro X

Naphthenic dielectric oil — Inhibited

PRODUCT DATA SHEET

Description

TENSIÓN CENTAURO oils are light, naphthenic oils, particularly recommended for use as insulating fluid in electrical equipment such as power and distribution transformers, circuit breakers, rheostats, etc. Their production process ensures the lack of any solid, polar or water compounds, as well as the lack of any corrosive sulphur, which has been proven to be extremely damaging to transformers.

Classification and standards

TENSIÓN CENTAURO X is an inhibited oil formulated with antioxidant additives. It complies with the IEC 60296 and ASTM D3487 Type II standards.

Technical specifications

Values according to ASTM D3487 standard.

Properties	Units	Method	Min.	Max.	Typical value
Physical properties					
Appearance		ASTM D1524		Bright and clear	
Colour		ASTM D1500		0.5	L0.5
Relative density at 15 °C	g/cm3	ASTM D1298		0.91	0.873
Viscosity at 0 °C	cSt	ASTM D445		76.0	67.29
Viscosity at 40 °C	cSt	ASTM D445		12.0	9.99
Viscosity at 100 °C	cSt	ASTM D445		3.0	2.48
Pour point	°C	ASTM D97		-40	-48
Aniline point	°C	ASTM D611	63		82
Interfacial tension	dynes/cm	ASTM D971	40		45
Chemical properties					
Acidity	mg KOH/g	ASTM D974		0.03	<0.01
Corrosive sulphur		ASTM D1275B		Non-corrosive	
Water content	ppm	ASTM D1533		35	22
Inhibitor content	% weight	ASTM D2668		0.3	0.3
Electrical properties					
DDF 60 Hz, 25 °C	%	ASTM D924		0.05	0.004
DDF 60 Hz, 100 °C	%	ASTM D924		0.30	0.142
Breakdown voltage 60 Hz - disk electrodes	kV	ASTM D877	30		56
BDV 60 Hz - VDE 0.04 in (1.02 mm) gap	kV	ASTM D1816	20		38
BDV 60 Hz - VDE 0.08 in (2.03 mm)	kV	ASTM D1816	35		

Properties	Units	Method	Min.	Max.	Typical value
gap					
BDV impulse, 25 °C, 1-in (25.4 mm) gap	kV	ASTM D3300	145		>300
Gassing tendency	uL/min	ASTM D2300		30	15.4
Oxidation stability — Acid-sludge test					
TAN (72 h)	mg KOH/g	ASTM D2440		0.3	<0.01
TAN (164 h)	mg KOH/g	ASTM D2440		0.4	<0.01
Sludge (72 h)	% weight	ASTM D2440		0.1	<0.01
Sludge (164 h)	% weight	ASTM D2440		0.2	<0.01
Rotating bomb test	min	ASTM D2112	195		295
Health, safety and the environment					
Flash point	°C	ASTM D92	145		154
PCB content	ppm	ASTM D4059		Undetectable	

Note: Typical values correspond to current production. Whilst future production will conform to Repsol's specification, variations in these characteristics may occur. The guaranteed minimum/maximum values define the product specification.